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LAPORAN TUGAS JOBSHEET 5

Latihan Praktikum

- Kode Program Mahasiswa15.java

```
package BruteForceDivideConquer.minggu5;
public class Mahasiswa15 {
    String nama;
    String nim;
    int tahunMasuk;
    int nilaiUTS;
    int nilaiUAS;

    Mahasiswa15(String nama, String nim, int tahunMasuk, int uts, int uas) {
        this.nama = nama;
        this.nim = nim;
        this.tahunMasuk = tahunMasuk;
        this.nilaiUTS = uts;
        this.nilaiUAS = uas;
    }
}
```

- Kode Program NilaiMahasiswa15.java

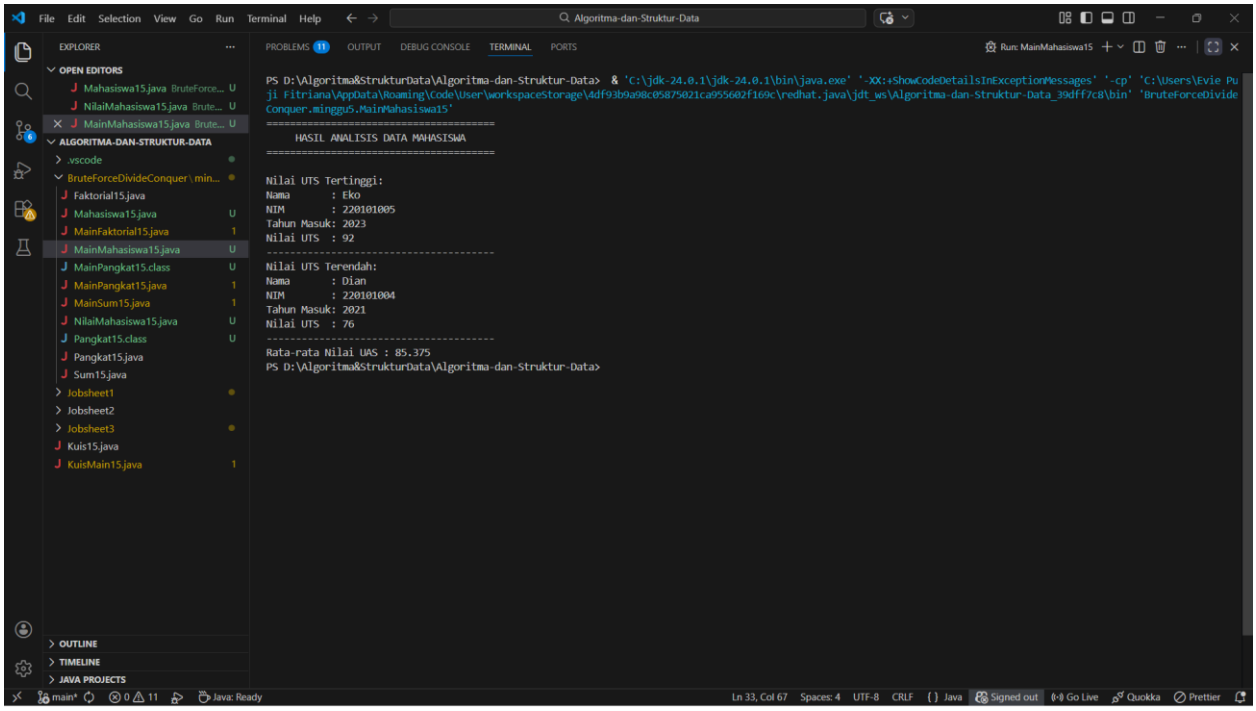
```
package BruteForceDivideConquer.minggu5;
public class NilaiMahasiswa15 {
    public static Mahasiswa15 utsTertinggi(Mahasiswa15[] mhs, int l, int r) {
        if (l == r) {
            return mhs[l];
```

```
    }
    int mid = (l + r) / 2;

    Mahasiswa15 kiri = utsTertinggi(mhs, l, mid);
    Mahasiswa15 kanan = utsTertinggi(mhs, mid + 1, r);

    if (kiri.nilaiUTS > kanan.nilaiUTS) {
        return kiri;
    } else {
        return kanan;
    }
}

public static Mahasiswa15 utsTerendah(Mahasiswa15[] mhs, int l, int r) {
    if (l == r) {
        return mhs[l];
    }
    int mid = (l + r) / 2;

    Mahasiswa15 kiri = utsTerendah(mhs, l, mid);
    Mahasiswa15 kanan = utsTerendah(mhs, mid + 1, r);

    if (kiri.nilaiUTS < kanan.nilaiUTS) {
        return kiri;
    } else {
        return kanan;
    }
}

public static double rataUAS(Mahasiswa15[] mhs) {
    double total = 0;
```

```
        for (int i = 0; i < mhs.length; i++) {
            total += mhs[i].nilaiUAS;
        }
        return (double) total / mhs.length;
    }
}
```

- Kode Program MainMahasiswa15.java

```
package BruteForceDivideConquer.minggu5;
public class MainMahasiswa15 {
    public static void main(String[] args) {
        Mahasiswa15[] data = {
            new Mahasiswa15("Ahmad", "220101001", 2022, 78, 82),
            new Mahasiswa15("Budi", "220101002", 2022, 85, 88),
            new Mahasiswa15("Cindy", "220101003", 2021, 90, 87),
            new Mahasiswa15("Dian", "220101004", 2021, 76, 79),
            new Mahasiswa15("Eko", "220101005", 2023, 92, 95),
            new Mahasiswa15("Fajar", "220101006", 2020, 88, 85),
            new Mahasiswa15("Gina", "220101007", 2023, 80, 83),
            new Mahasiswa15("Hadi", "220101008", 2020, 82, 84)
        };
        Mahasiswa15 tertinggi = NilaiMahasiswa15.utsTertinggi(data, 0, data.length - 1);
        Mahasiswa15 terendah = NilaiMahasiswa15.utsTerendah(data, 0, data.length - 1);
        double rata = NilaiMahasiswa15.rataUAS(data);

        System.out.println("========================================");
        System.out.println("     HASIL ANALISIS DATA MAHASISWA     ");
        System.out.println("========================================");
        System.out.println("\nNilai UTS Tertinggi: ");
        System.out.println("Nama       : " + tertinggi.nama);
```

```
        System.out.println("NIM        : " + tertinggi.nim);
        System.out.println("Tahun Masuk: " + tertinggi.tahunMasuk);
        System.out.println("Nilai UTS  : " + tertinggi.nilaiUTS);
        System.out.println("--------------------------------------");
        System.out.println("Nilai UTS Terendah: ");
        System.out.println("Nama       : " + terendah.nama);
        System.out.println("NIM        : " + terendah.nim);
        System.out.println("Tahun Masuk: " + terendah.tahunMasuk);
        System.out.println("Nilai UTS  : " + terendah.nilaiUTS);
        System.out.println("--------------------------------------");
        System.out.println("Rata-rata Nilai UAS : " + rata);
    }
}
```

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